Indian Institute Of Information Technology – Vadodara

Embedded System Project Summary

Group – 9:

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Embedded system for Nutrient Stress detection

System Integration

The project consists of two main modules: the embedded hardware system and the machine learning model.

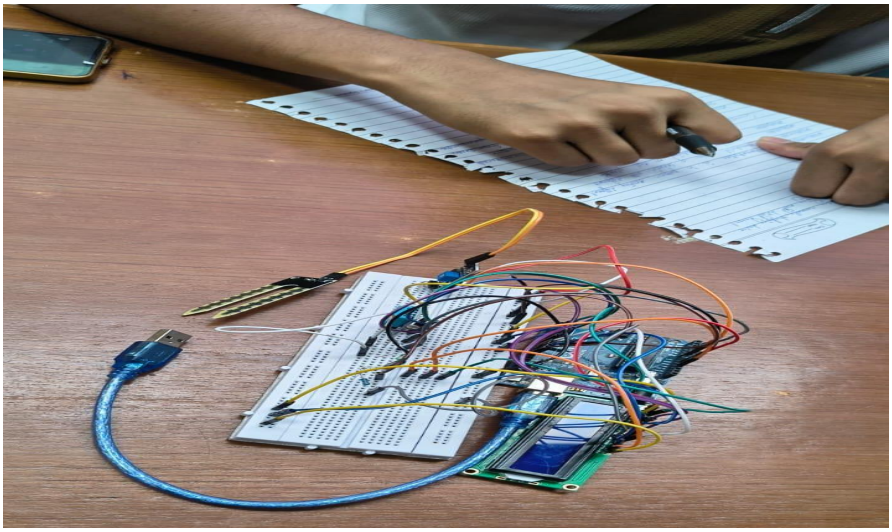
The embedded system collects real-time environmental data such as soil moisture, temperature, humidity, and light intensity using sensors connected to the Arduino.

The machine learning model is trained using soil nutrient data (N, P, K, pH) to classify soil condition.

The proposed integration flow is:

Sensors → Arduino → Data Processing → ML Model → Output (LCD/Alert)

This integration will allow the system to provide more accurate and intelligent nutrient stress detection in real-time.



Testing and Validation

The system has been tested in two stages:

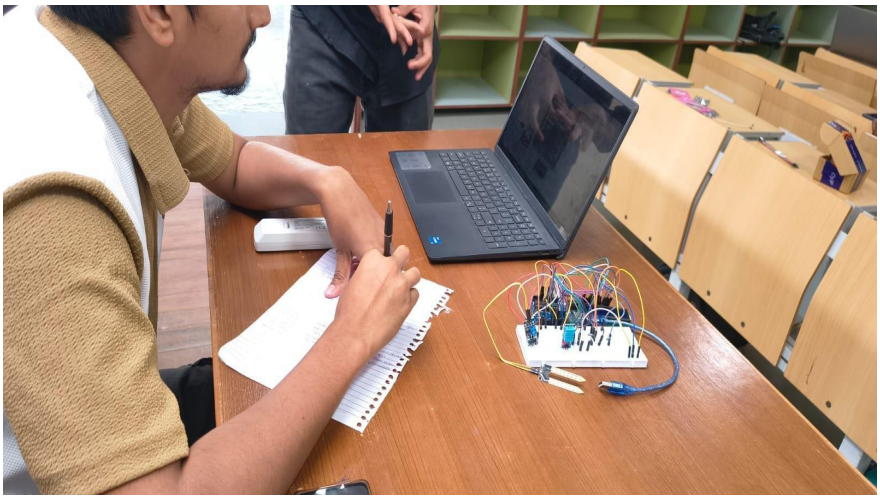
Hardware Testing:

- Circuit verified using Tinkercad simulation
- LCD output and LED indication tested
- Sensor readings validated under different conditions

ML Model Testing:

- Model trained on soil dataset
- Tested using sample inputs
- Verified outputs for Normal, Deficient, and Stress conditions

Both modules are working independently and ready for integration.

Performance and Observations

- Threshold-based system provides basic stress detection
- CNN-LSTM model improves prediction capability
- System is low-cost and easy to implement
- Model successfully classifies soil condition based on input data
- Approach is scalable for smart agriculture application.